

AI, Data Modelling & Ethics

Extracting Safety Intelligence from Healthcare Text

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About Me

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Developing AI methods for healthcare safety that are **effective**, **equitable**, and **ethical**.

Healthcare AI research spanning:

- Clinical text & reports
- Electronic health records
 - Physiological sensors
- Multimodal data fusion

Funded by NHSX, The Health Foundation,
NIHR

Our Partner: HSIB → HSSIB

Healthcare Safety Investigation Branch (HSIB)

Established in 2017 to improve patient safety through independent investigations. We collaborated with HSIB on this research, analysing their maternity investigation reports.

Transformation in October 2023:

HSIB transformed into two separate organisations:

- **MNSI** — Maternity and Newborn Safety Investigations
- **HSSIB** — Health Services Safety Investigations Body

Core functions:

- Conduct independent patient safety investigations across NHS England
- Publish reports with safety recommendations
- Deliver patient safety investigation training

Investigations focus on learning, not blame. Staff who contribute are legally protected, encouraging open and honest reporting.

What We'll Explore Today

1. The Challenge: How do we balance using data with protecting people?

2. The Methods: Can AI find patterns in safety data reliably and fairly?

3. The Ethics: What frameworks ensure AI systems remain accountable?

Using UK maternity safety as our case study

01

The Challenge

Unlocking safety insights from unstructured data

The Challenge

Making sense of safety data at scale

The problem:

- Healthcare generates vast **unstructured text**
- Safety reports contain actionable insights
- Patterns are **hard to spot** at scale
- Manual analysis cannot keep pace

By identifying recurring themes, we can help prevent future harm.

The considerations:

- Health data requires careful handling under GDPR
- AI must perform **fairly across all groups**
- Findings must be **traceable** and verifiable
- Human oversight remains essential

Responsible AI means designing in safeguards from the start.

Can AI help us learn from safety incidents — reliably, fairly, and responsibly?

*How can we use AI to extract
actionable safety intelligence
from hundreds of
unstructured reports
— responsibly and fairly?*

02

The Methods

Applying AI to UK maternity safety

Method 1: AI Concept Annotation

The I-SIRch Framework

Approach:

- ML model trained on expert annotations
- Human factors taxonomy (27 concepts)
- Human-in-the-loop validation

Key innovation:

Focus on *human factors* — communication, teamwork, decision-making.

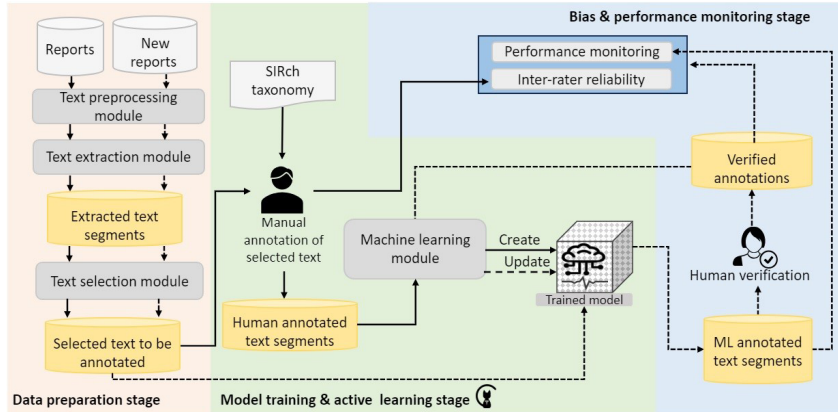
Top concepts in HSIB reports:

Communication	159
Teamworking	155
Assessment	150
Physical factors	118

Communication issues found in **159 of 188** reports — the most prevalent human factor.

I-SIRch: Framework

Human factors + AI = Enhanced patient safety intelligence



The SIRch Taxonomy: SEIPS Framework

27 concepts enable comprehensive analysis of socio-technical factors in maternity care.

SEIPS Framework Components:

1. External Environment (5)

Policy, Societal, Economic, COVID, Geographical

2. Internal Environment (4)

Physical layout, Acuity, Availability, Time of day

3. Organisation (8)

Team culture, Teamworking, Communication,
Documentation, Escalation/referral, Guidance,
Language barrier

4. Job/Task (7)

Assessment, Care planning, Dispensing, Monitoring,
Risk assessment, Situation awareness, Obstetric
review

5. Technologies & Tools (2)

Issues, Interpretation (e.g. CTG)

6. Person (6 each)

Patient & Staff characteristics: Physical,
Psychological, Language, Disability, Training,
Attendance

Performance: Slips/lapses, Decision errors, Rule
breaking

SEIPS = Systems Engineering Initiative
for Patient Safety

Analyses how work system design influences patient safety outcomes.

I-SIRch: Training the Model

Expert annotation with human-in-the-loop validation

Human annotation process:

1. Three domain experts annotated reports
2. Used SIRch taxonomy (27 human factors)
3. **Ethnicity blinded** during annotation
4. Inter-rater reliability: **80.15%**

Blinding ethnicity during human annotation prevents biases entering the training data.

Dataset composition:

188

Real investigation reports
(2019–2022, 3,122 sentences)
Ethnicity: 76 reports

Training: 91 reports (1,162 sentences)
Testing: 97 reports (1,960 sentences)

Key principle: Train on real data, test on real unseen data to validate real-world performance.

I-SIRch: Fairness Across Ethnic Groups

Does the model perform equally for all groups?

Performance by ethnicity:

Ethnic Group	Bal. Acc.	Recall	n
White British	84%	67%	688
Asian	84%	68%	87
Black	82%	65%	81
Mixed Background	85%	69%	13
Other White	82%	64%	46

n = number of concept annotations tested

Precision was 1.0 (std=0) across all groups.

Statistical analysis:

Wilcoxon signed-rank test:

$Z = -0.806$, $p = 0.42$

No significant difference between Black and White British groups.

The model performs **consistently** across all ethnic groups — no evidence of bias.

AI systems must be tested for fairness across protected characteristics. ✓

Why Metrics Matter

Evaluating AI fairly in healthcare

The problem with accuracy alone:

In imbalanced datasets, accuracy can mislead.

Example: If 95% of cases are “safe”, predicting everything as “safe” achieves 95% accuracy — but misses every incident.

Missing critical cases is unacceptable.

Key metrics for equitable AI:

Recall — Did we find all relevant cases?

Precision — Are predictions correct?

Balanced Acc. — Fair across groups

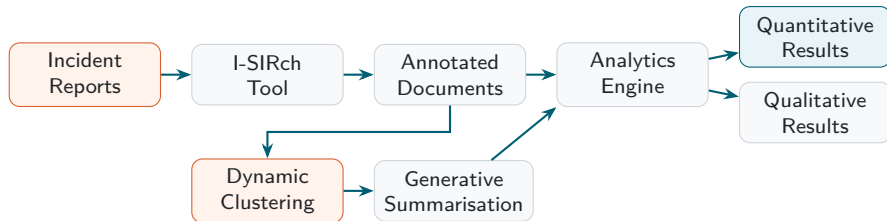
F-score — Balances precision & recall

Test across demographic groups to detect bias.

ICO Principle: AI systems must be tested for fairness across protected characteristics. ✓

Method 2: Multi-Document Summarisation

I-SIRch:CS Framework



Technical approach:

Sentence embeddings (all-MiniLM-L6-v2) →
K-means clustering → Abstractive
summarisation with **BART**

Offline models ensure GDPR compliance

Best model (BART) performance:

Relevance: **0.922** | Coherence: **0.794** |
Diversity: **0.806**

Dataset: 188 reports, 3,760 annotated
sentences across 27 concepts

Summarisation Dataset Composition

3,760 annotated sentences across 27 human factors concepts

ID	Concept	Index	Label
1	Acuity (54)	2	Antenatal (69)
3	Assessment, investigation, testing, screening (381)	4	COVID (142)
5	Care Planning (132)	6	Communication (477)
7	Decision error (169)	8	Dispensing, administering (62)
9	Documentation (168)	10	Escalation/referral factor (158)
11	Guidance factor (42)	12	Interpretation (197)
13	Language barrier (30)	14	Local guidance (88)
15	Monitoring (118)	16	National and local guidance (80)
17	National guidance (169)	18	Obstetric review (147)
19	Physical characteristics (320)	20	Physical layout and Environment (35)
21	Psychological characteristics (54)	22	Risk assessment (94)
23	Situation awareness (77)	24	Slip or lapse (188)
25	Teamworking (530)	26	Technologies and Tools-issues (112)
27	Training and education (40)		

Evaluating Equity in Summarisation

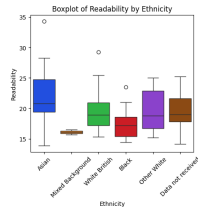
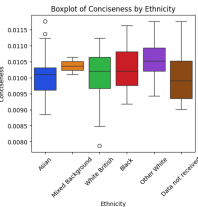
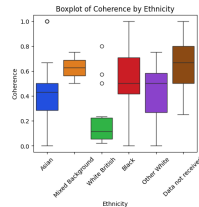
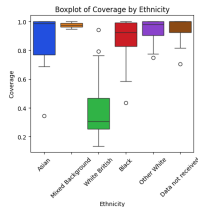
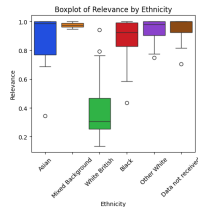
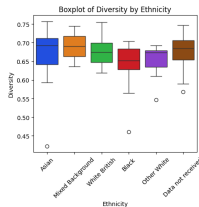
Do summaries perform fairly across ethnic groups?

BART performance by ethnicity:

Group	Rel.	Coh.	Div.
Asian	0.89	0.46	0.67
Black	0.87	0.54	0.64
Mixed	0.97	0.63	0.69
Other White	0.93	0.42	0.65
White British	0.40	0.18	0.68

Ethnicity data: 76 of 188 reports (Batch 1)

Lower White British scores attributed to larger sample size introducing variability, not model bias.



ICO Principle: Fairness, transparency, and accountability must be designed in from the start. ✓

How We Evaluate Summary Quality

Six metrics assess different quality dimensions

- **Diversity:** Variety of vocabulary (unique words / total words)
- **Relevance:** Similarity to original text (cosine similarity)
- **Coverage:** Proportion of key concepts captured
- **Coherence:** Logical flow between sentences
- **Conciseness:** Brevity of summary
- **Readability:** Ease of understanding (Flesch score)

Higher values indicate better performance. BART achieves: Diversity 0.806 | Relevance 0.922 | Coherence 0.794

These metrics enable objective comparison across models and demographic groups

Example Summaries Generated by BART

Demonstrating summarisation quality and traceability

Communication concept:

Heading: Staff not heard

Staff voiced concerns to the senior obstetrician but were left feeling unheard. No formal debriefing occurred afterwards, which staff valued. The opportunity to share reflections wasn't completed with all involved. Incremental delays and communication breakdown further impacted the DDI.

Acuity concept:

Heading: Reviews with seniors did not occur

The ambulance Trust experienced high call volumes. Due to high labour ward acuity, initial decisions weren't discussed with senior clinicians. Senior face-to-face review didn't occur until 16:05, 2 hours 50 minutes after recognising abnormalities. The abnormal CTG wasn't reviewed by the senior obstetrician.

Each summary is traceable to source file and sentence IDs, enabling verification and accountability.

Traceability in Summarisation

Linking summary content back to source documents for transparency and auditability

Ethnic Group	Outcome	Summary	Related IDs	Sentence
Asian	Cooled baby	High workload prevented timely review of abnormal CTGs and recognition of risk factors, delaying intervention and impacting cooled baby outcomes.	370, 373, 374	
Asian	Early NND	Delays in post-delivery assessments of a baby prevented recognition of condition and need for monitoring, impacting an early NND.		372
Asian	Stillbirth	Incomplete maternal observations and assessments contributed to lack of recognition of clinical deterioration prior to stillbirths.	366, 367, 368	
Black	Cooled baby	Lack of holistic assessments meant missed screening for gestational diabetes and growth scans, potentially impacting cooled baby outcomes.	289, 290, 291, 292	
Black	MD	Lack of pregnancy test and referral to additional support represented missed care opportunities prior to a maternal death.		377, 378
Data not received	MD	Gaps in assessments and documentation, and lack of key lab tests represented missed opportunities prior to maternal deaths.	379, 380, 381	
Mixed Background	Cooled baby	Lack of testing for Zika virus prevented full risk assessment for a cooled baby.		371

Ethical Risks of AI Summarisation

Identified risks and mitigations

Risk	Mitigation
Information hallucination & bias amplification	Diversify training datasets; implement debiasing techniques; cross-reference with original texts; regular bias audits
Inadequate content control	Develop models with embedded ethical constraints; employ sentence ID traceability for verification
Sensitive data exposure	Use offline LLMs to reduce breach risks; balance security against performance needs
Sample size variability	Robust data preprocessing; ML algorithms customised for high variability datasets

Key safeguard: Sentence-level traceability enables verification of every summary, ensuring verifiability, trustworthiness, and accountability.

*We can build
powerful AI systems.*

***Should** we?*

*And **how** do we do so
responsibly?*

03

The Ethics

Data protection and responsible AI

The Ethical Tension

The Promise

AI can identify patterns
at scale that inform safety
improvements

VS

The Risk

AI can amplify bias and
violate individual rights

The legal and ethical frameworks exist to navigate this tension.

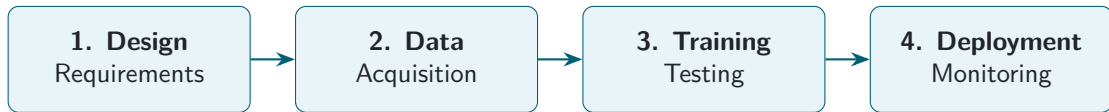
UK GDPR • ICO AI Toolkit • Individual Rights • Human Oversight

UK GDPR: Key Articles for AI

Article	Title	Key requirement
Art. 5	Principles	Lawfulness, fairness, transparency, purpose limitation, data minimisation
Art. 6	Lawful basis	Must have valid legal grounds for processing personal data
Art. 9	Special categories	Health data requires explicit consent or public interest grounds
Art. 15	Right of access	Individuals can request a copy of their data and processing information
Art. 17	Right to erasure	Individuals can request deletion of their personal data
Art. 22	Automated decisions	Right not to be subject to solely automated decisions with legal effects

AI systems processing health data must address all of these requirements.

ICO AI & Data Protection Framework



Fairness

Bias, discrimination, adverse effects

Transparency

Explainability, traceability

Accountability

DPIA, roles, oversight

Each stage requires specific controls — risks change as the system evolves

ICO Risk Toolkit: Key Areas

Risk area	What the ICO requires
Accountability	Assign clear roles; conduct DPIA before deployment; document decisions
Lawfulness	Identify and document lawful basis; meet Article 9 conditions for health data
Fairness	Assess bias risks; consider impact on individuals; engage affected groups
Transparency	Explain AI use to data subjects; enable meaningful human review
Security	Protect training data; secure model from adversarial attacks
Data minimisation	Use only necessary data; consider synthetic or anonymised alternatives
Individual rights	Facilitate access, rectification, erasure throughout AI lifecycle

Source: ICO AI and Data Protection Risk Toolkit

Meaningful Human Review

Article 22 — Automated decision-making

ICO Requirements:

1. Humans must **check** AI recommendations
2. Reviewers must have **actual influence**
3. Must **weigh up** and interpret outputs

Not a “token gesture” — genuine authority to override.

In this research:

Human-in-the-loop validation
Expert annotation review
Traceability to sources
Domain expert interpretation

AI assists, humans decide

Patient & Public Involvement

PPI in this research:

PPI contributors reviewed and provided input into the human factors taxonomy used for annotation.

Ongoing commitment:

Continued PPI engagement throughout research, particularly with affected communities.

Why PPI matters:

- Aligns with patient needs
- Increases transparency
- Identifies biases from lived experience
- Builds trust

How do we ensure AI tools serve everyone equitably?

An Open Challenge

*“We invest enormous effort teaching machines to learn from data.
But what happens when they need to **forget**?”*

The right to erasure meets the reality of machine learning.

An open challenge for ethical AI.

The Unlearning Problem

Article 17 — Right to erasure

The fundamental tension:

Machine Learning
Models learn from data

↕ **Conflict?**

Right to Erasure
Individuals can request deletion

The challenge:

Once a model has learned from data, how do you **remove that learning**?

Current approaches:

- Retrain from scratch (expensive)
- Approximate unlearning (imperfect)
- Data provenance tracking (complex)

Most healthcare AI has **no mechanism** to comply with erasure requests — an unresolved legal risk.

*If someone asked today:
“What data do you hold on me,
and how is AI using it?”
Could you answer?*

Thank You

References:

- Cosma et al. — (2024) I-SIRch: AI-powered concept annotation tool for equitable extraction and analysis of safety insights from maternity investigations, *International Journal of Population Data Science*, 9(2). doi: 10.23889/ijpds.v9i2.2439.
- Cosma et al. — (2024). Intelligent Multi-document Summarisation for Extracting Insights on Racial Inequalities from Maternity Incident Investigation Reports. In: Xie, X., Styles, I., Powathil, G., Ceccarelli, M. (eds) *Artificial Intelligence in Healthcare. AliH 2024. Lecture Notes in Computer Science*, vol 14976. Springer, Cham. https://doi.org/10.1007/978-3-031-67285-9_23